

Building a Multi-Cluster Service Mesh from the Ground Up with Kilo

CozySummit 2026
virtual

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FAQ

- why do we need *another* multi-cluster service mesh?
- how do we automate cluster peering?
- why not just load-balancer services?
- isn't kubernetes basically a service-mesh?

■ Let's Define Some Terms

■ What's a Service Mesh?

- enable service connectivity
 - abstract networking
 - abstract discovery
- provide features you'd have to re-implement
 - encryption
 - load-balancing
 - shadowing
 - etc

■ Let's Define Some Terms

▣ What's a Service Mesh?

| why would you even want that?

■ Let's Define Some Terms

▣ What's a Service Mesh?

| if they're really so great, name some

■ Let's Define Some Terms

▣ What's a Service Mesh?

- Linkerd
- Istio
- Cilium
- . . .



What's a *Multi-Cluster* Service Mesh?

it provides the same features but across cluster boundaries

■ Let's Define Some Terms

■ What's a *Multi-Cluster* Service Mesh?

| ok, so why would you want *that*?



What Are We Building?

■ Let's Define Some Terms

▣ What Are We Building?

...the hand-wavy part

■ Let's Define Some Terms

■ What Are We Building?

| are we really building all of that?
| from the ground up?
no, i make the rules!

we're going to focus on:

- enable service connectivity
 - abstract networking
 - abstract discovery
- ~~provide features you'd have to re-implement~~
 - encryption
 - load balancing
 - shadowing
 - etc

The Plan

- enable service connectivity
 - abstract networking
 - abstract discovery

The Rest of the Owl

How to draw an owl

1.



1. Draw some circles

2.



2. Draw the rest of the fucking owl

- enable service connectivity
 - abstract networking
 - full routability within clusters
 - pod-to-pod connectivity
 - pod-to-host connectivity
 - pod-to-service connectivity
 - full routability between clusters
 - cluster-to-cluster connectivity
 - abstract discovery
 - enable local service discovery
 - enable remote service discovery

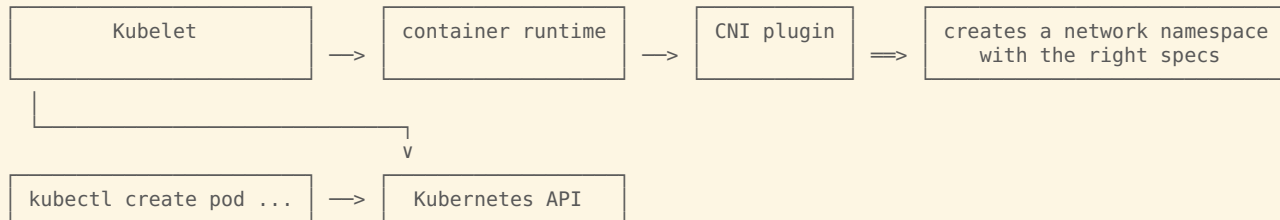
■ From the Ground Up

■ Pod-to-Pod Connectivity

the laws of Kubernetes networking

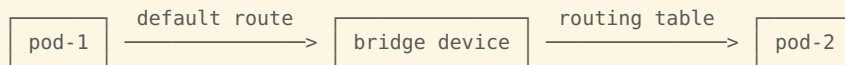
1. every pod needs to be able to talk to every other pod
2. ... that's pretty much it

| how do we implement this?
CNI



 Pod-to-Pod Connectivity

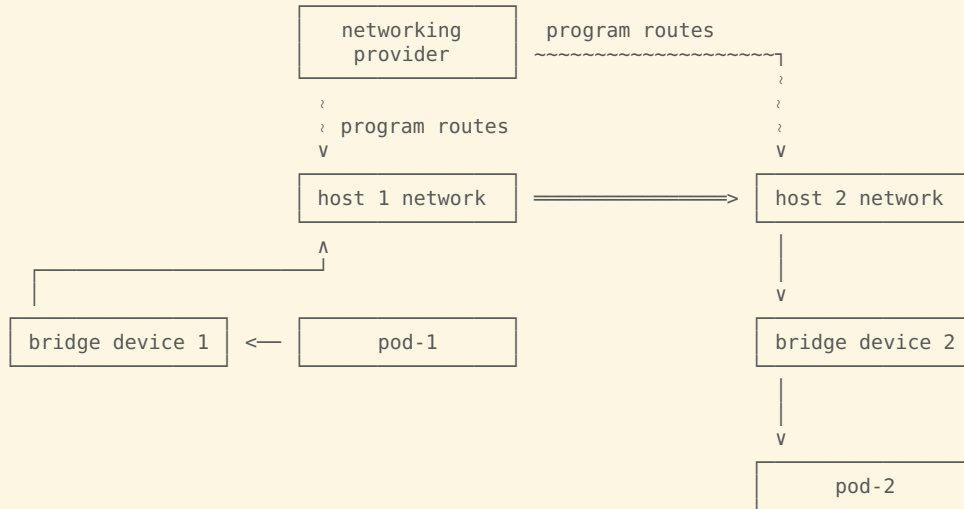
```
{
  "cniVersion": "0.4.0",
  "name": "a-basic-cni-network",
  "plugins": [
    {
      "name": "kubernetes",
      "type": "bridge",
      "bridge": "kube-bridge",
      "isDefaultGateway": true,
      "forceAddress": true,
      "mtu": 1500,
      "ipam": {
        "type": "host-local"
      }
    },
    {
      "type": "portmap",
      "snat": true,
      "capabilities": {
        "portMappings": true
      }
    }
  ]
}
```



| cool, what about a pod on a different node?



we need a networking provider
there's like, a million of them



■ From the Ground Up

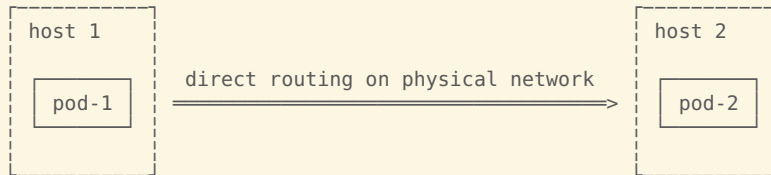
■ Pod-to-Pod Connectivity

■ Let's Use Kilo

■ From the Ground Up

▒ Pod-to-Pod Connectivity

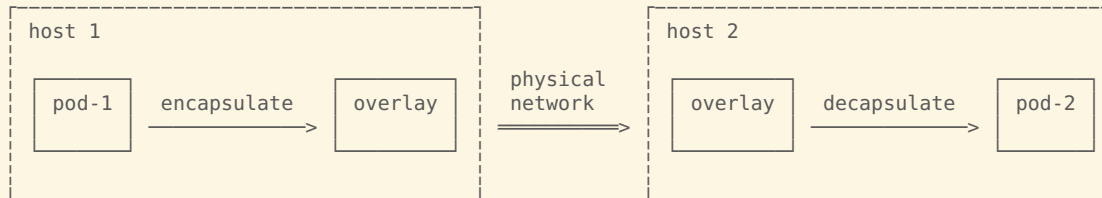
■ Let's Use Kilo



■ From the Ground Up

▒ Pod-to-Pod Connectivity

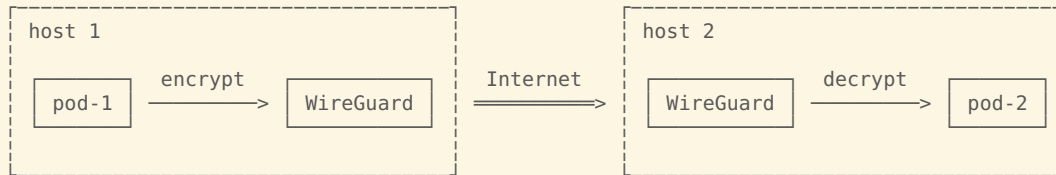
■ Let's Use Kilo



■ From the Ground Up

■ Pod-to-Pod Connectivity

■ Let's Use Kilo

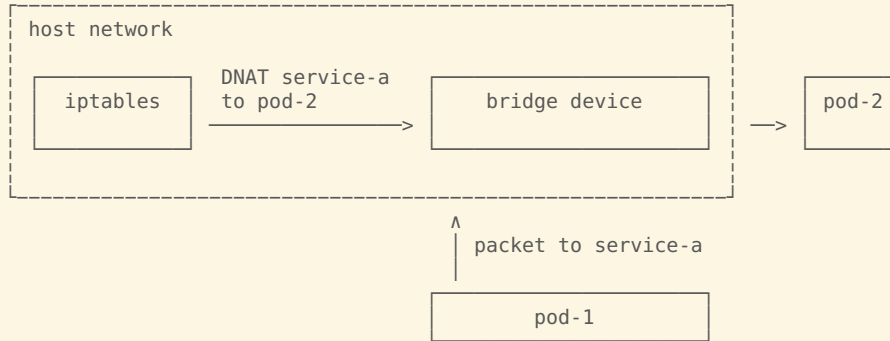


■ Pod-to-Service Connectivity

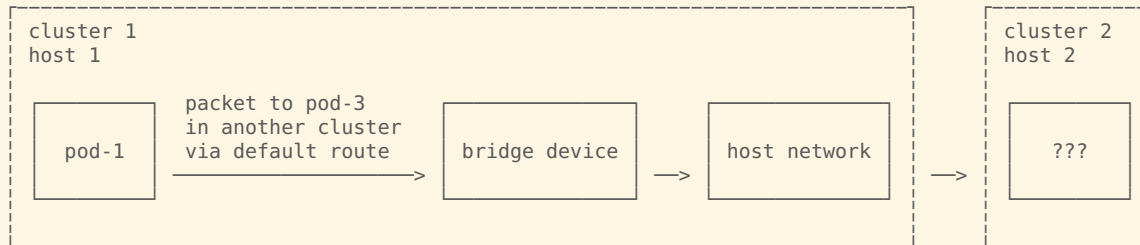
| what even is a service?

services are just a convenient abstraction providing a stable network identity for a set of variable pod ips

kube-proxy



■ Cluster-to-Cluster Connectivity



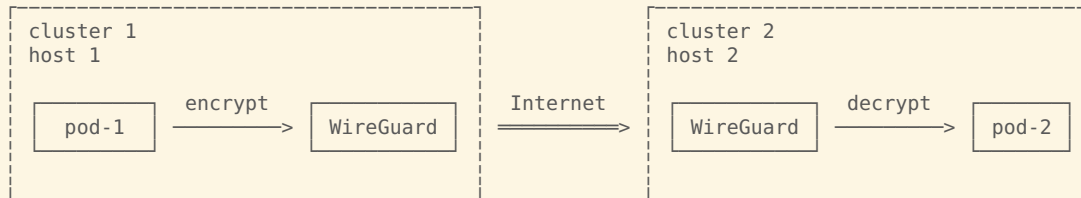
look familiar?

this is the same problem as pod-to-pod connectivity

requirements

- discover routes
- secure underlying network
- the rest of the owl!

we need a networking provider



■ Local Service Discovery

■ From the Ground Up

■ Local Service Discovery

■ DNS

■ From the Ground Up

■ Local Service Discovery

■ DNS

| why?

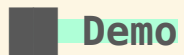
| why not?

■ Remote Service Discovery

enter multi-cluster services api

github.com/squat/service-reflector (
<https://github.com/squat/service-reflector>)

- enable service connectivity
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 - enable local service discovery
 - enable remote service discovery



 Create Clusters

```
# create clusters
kind create cluster --config kind-1.yaml --kubeconfig k1
kind create cluster --config kind-2.yaml --kubeconfig k2
```

snippet +exec is disabled, run with -x to enable

Install Kilo

```
# install kilo
kubectl --kubeconfig k1 create secret generic kubeconfig --from-file=kubeconfig=k1 -n kube-system
kubectl --kubeconfig k2 create secret generic kubeconfig --from-file=kubeconfig=k2 -n kube-system

kubectl --kubeconfig k1 apply -f
https://raw.githubusercontent.com/squat/kilo/refs/heads/main/manifests/crds.yaml
kubectl --kubeconfig k2 apply -f
https://raw.githubusercontent.com/squat/kilo/refs/heads/main/manifests/crds.yaml

kubectl --kubeconfig k1 apply -f kilo-kind-userspace.yaml
kubectl --kubeconfig k2 apply -f kilo-kind-userspace.yaml

# patch k2 to give it a different kilo CIDR
kubectl --kubeconfig k2 patch ds -n kube-system kilo -p '{"spec":
{"template":{"spec":{"containers":[{"name":"kilo","args":["--hostname=$(NODE_NAME)","--create-interface=false",
"--kubeconfig=/etc/kubernetes/kubeconfig","--mesh-granularity=full","--subnet=10.6.0.0/16"]}]}}}}'

# ensure cluster is stable
kubectl --kubeconfig k1 wait nodes --all --for=condition=Ready
kubectl --kubeconfig k2 wait nodes --all --for=condition=Ready
```

snippet +exec is disabled, run with -x to enable

 Install Webcam Service

```
# install webcam service in k2
kubectl --kubeconfig k2 apply -f
https://raw.githubusercontent.com/squat/generic-device-plugin/main/manifests/generic-device-plugin.yaml
kubectl --kubeconfig k2 apply -f mjpeg.yaml
kubectl --kubeconfig k2 wait --for=condition=Ready pod/mjpeg
```

snippet +exec is disabled, run with -x to enable

 Generate Image from Webcam Service

```
# connect to service
PORT_FORWARD_OUTPUT="$(kubectl --kubeconfig k2 port-forward svc/mjpeg http & echo $!) | (sed '/Forwarding
from/q' ; cat > /dev/null &)"
PORT_FORWARD_PID="$(echo "$PORT_FORWARD_OUTPUT" | sed -n 1p)"
PORT_FORWARD_HOST="$(echo "$PORT_FORWARD_OUTPUT" | sed -n 's/Forwarding from \(.*\) -> 8080/\1/p')"
trap 'kill "$PORT_FORWARD_PID"' EXIT

# get jpeg
curl -s "http://$PORT_FORWARD_HOST/jpeg" | chafa --format symbols --size 50x50
kill "$PORT_FORWARD_PID"
trap - EXIT
```

snippet +exec is disabled, run with -x to enable

 Peer Clusters

```
# Register the nodes in cluster1 as peers of cluster2.
for n in $(kubectl --kubeconfig k1 get no -o name | cut -d '/' -f2); do
    # Specify the service CIDR as an extra IP range that should be routable.
    kgctl --kubeconfig k1 showconf node "$n" --as-peer -o yaml --allowed-ips 10.43.0.0/16 | kubectl
--kubeconfig k2 apply -f -
done
# Register the nodes in cluster2 as peers of cluster1.
for n in $(kubectl --kubeconfig k2 get no -o name | cut -d '/' -f2); do
    # Specify the service CIDR as an extra IP range that should be routable.
    kgctl --kubeconfig k2 showconf node "$n" --as-peer -o yaml --allowed-ips 10.45.0.0/16 | kubectl
--kubeconfig k1 apply -f -
done
```

snippet +exec is disabled, run with -x to enable

 Generate Image from Peered Cluster Using Remote Service

```
(  
  docker exec 1-control-plane apt update  
  docker exec 1-control-plane apt install --yes dnsutils  
) > /dev/null 2>&1  
docker exec 1-control-plane bash -c 'curl "http://$(dig mjpeg.default.svc.cluster.local @10.45.0.10  
+short):8080/jpeg" --output - 2>/dev/null' | chafa --format symbols --size 50x50
```

snippet +exec is disabled, run with -x to enable

 Generate Image from Peered Cluster Using Local Service

```
(  
  docker exec 1-control-plane apt update  
  docker exec 1-control-plane apt install --yes dnsutils  
) > /dev/null 2>&1  
docker exec 1-control-plane bash -c 'curl "http://$(dig mjpeg.default.svc.cluster.local @10.43.0.10  
+short):8080/jpeg" --output - 2>/dev/null' | chafa --format symbols --size 50x50
```

snippet +exec is disabled, run with -x to enable

this fails!

Manually Mirror Service

```
cat <<EOF | kubectl --kubeconfig k1 apply -f -
apiVersion: v1
kind: Service
metadata:
  name: mjpeg
spec:
  ports:
    - name: http
      port: 8080
      protocol: TCP
      targetPort: http
  ---
apiVersion: v1
kind: Endpoints
metadata:
  name: mjpeg
subsets:
  - addresses:
    - ip: $(kubectl --kubeconfig k2 get pod mjpeg -o jsonpath='{.status.podIP}')
    ports:
    - name: http
      port: 8080
      protocol: TCP
EOF
```

snippet +exec is disabled, run with -x to enable

 Generate Image from Peered Cluster Using Local Service

```
(  
  docker exec 1-control-plane apt update  
  docker exec 1-control-plane apt install --yes dnsutils  
) > /dev/null 2>&1  
docker exec 1-control-plane bash -c 'curl "http://$(dig mjpeg.default.svc.cluster.local @10.43.0.10  
+short):8080/jpeg" --output - 2>/dev/null' | chafa --format symbols --size 50x50
```

snippet +exec is disabled, run with -x to enable

this works now!

 Generate Image from Peered Cluster Using Kubernetes Service Discovery

```
docker exec 1-control-plane curl "http://$(kubectl --kubeconfig k1 get service mjpeg -o jsonpath='{.spec.clusterIP}'):8080/jpeg" --output - 2>/dev/null | chafa --format symbols --size 50x50
```

snippet +exec is disabled, run with -x to enable

Manually Mirrored Endpoints Are Static and Fragile

```
kubectl --kubeconfig k2 delete pod mjpeg
kubectl --kubeconfig k2 apply -f mjpeg.yaml
docker exec l-control-plane curl "http://$(kubectl --kubeconfig k1 get service mjpeg -o jsonpath='{.spec.clusterIP}'):8080/jpeg" --output - 2>/dev/null | chafa --format symbols --size 50x50
```

snippet +exec is disabled, run with -x to enable

restarting the pod rolls its IP and breaks service discovery on the peered cluster

We Need a Control Loop

github.com/squat/service-reflector (
<https://github.com/squat/service-reflector>)

 **Thank You!**

FAQ

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- how do we automate cluster peering?
- why not just load-balancer services?
- isn't kubernetes basically a service-mesh?